How Much Can Conjunction Message Histories Predict Beyond Persistence?

Arjun Vijay Prakash
City Montessori School, Kanpur Road Campus
Lucknow, India

***Abstract*—Can a cutoff-safe history of Conjunction Data Messages predict a later reported $\log_{10} P_c$ better than copying today's report? On the public ESA Collision Avoidance Challenge archive the apparent mean-error win is real, but it is almost entirely a later collapse to the dataset floor of -30, not a typical-event accuracy gain and not a high-risk decision gain. Predictive information beyond persistence is largest 72 hours before closest approach and is consistent with zero at 12 hours on a matched event cohort. Unguarded XGBoost lowers MAE from 5.080 to 2.809 on 1659 held-out events, yet floor-excluded MAE rises from 4.073 to 7.562 (Wilcoxon $p = 0.91$). A validation-selected two-part floor hurdle reaches MAE 2.109 with $F_2 = 0$. On ESA's frozen official test, last-risk persistence matches Uriot $L = 0.694$; unguarded XGBoost explodes to $L = 1.75 \times 10^6$ after one overconfident high-risk miss. This is a research measurement, not flight software.**

***Index Terms*—conjunction data message, reported collision probability, forecast horizon, persistence, selective prediction**

I. Introduction

Two catalogued objects can pass close enough to generate a Conjunction Data Message (CDM) [1]. The reported collision probability P_c is recomputed as later tracking shrinks—or fails to shrink—the combined covariance [2]. Waiting usually improves the estimate and consumes planning time. ESA posed the forecasting form of this problem in the 2019 Collision Avoidance Challenge: predict the final reported $\log_{10} P_c$ from messages with time to closest approach of at least two days [3], [4]. Persistence—copying the latest pre-cutoff report—was already strong. Only 12 of 96 teams beat it on ESA-style loss. Conjunction volume has since grown with mega-constellations [5], [6], so the same question is more, not less, relevant: how much extra information sits in an early CDM history?

Physics-based P_c forecasting via expected covariance reduction [7] and operational work on probability dilution [2], [8] suggest that large early covariances can make P_c look safely small, after which later updates either collapse the report to negligible or concentrate it. A recent model predicts which conjunctions will keep high covariance, for extra tracking [9]. The target here is different: the later *reported* $\log_{10} P_c$ under a frozen information cutoff. That target is itself an estimate, not a ground-truth collision occurrence.

This paper makes three contributions. (1) A leakage-safe measurement of extra information beyond persistence as a function of forecast horizon. (2) A decomposition showing that mean-absolute-error (MAE) gains are dominated by collapses to the archive floor -30. (3) Evidence that mean-error optimization can worsen ESA-style high-risk loss. The working hypotheses, and a one-line preview, are: H1 extra information decays toward closest approach (supported on a matched cohort); H2 most of the MAE win is floor jumps, not typical-event accuracy (supported); H3 extra information does not improve the $\log_{10} P_c \geq -6$ tail under this protocol (supported); H4 covariance volume and the max-risk gap are associated with later movement and floor membership (mixed after partialling).

II. Dataset and Protocol

Labeled CDMs from the ESA training archive comprise 162,634 rows and 13,154 events, 2015–2019, anonymized [4]. An event needs a message at or before the cutoff and a later labeled update, yielding 8,293 eligible events at 48 hours. Train, validation, calibration, and test are event-disjoint (3,731 / 1,659 / 1,244 / 1,659). Seed 42 is the reported local split. Five grouped redraws use seeds 42–46 with no test-tuned knobs. Official-test inputs are public Kelvins `test_data.csv`; labels are the released `true_risk` column from Zenodo 4463683, joined after freeze ($n = 2167$). No hyperparameter was changed after seeing official-test scores. The public file has no calendar dates and no object-pair identifiers, so a year-based split and NORAD grouping are impossible; a mission-grouped split and the official test are the available robustness checks. Fig. 1 records the counts.

Let risk be the latest pre-cutoff reported $\log_{10} P_c$ and y the later reported value. Persistence copies risk. A residual model predicts $\Delta = y - \mathrm{risk}$ and reconstructs $\widehat{y} = \mathrm{risk} + \widehat{\Delta}$. An event is a floor event if $y \leq -30 + 10^{-6}$. That bound is a reporting floor / censored target, not a precise physical P_c. Confirmatory analyses are H1–H4 on the frozen split. Horizon, threshold, and ablation sweeps are exploratory; they are not multiplicity-corrected.

Features are cutoff-safe snapshot fields, history summaries of the same fields, and covariance trends (234 numeric columns after encoding; 77 snapshot, 68 history, 88 covariance trends). The shipped regressor excludes `dilution_gap` $=$ `max_risk_estimate` $-$ `risk`, which is reserved as an H4 probe. XGBoost uses squared error, 250 trees, learning rate 0.05, depth 4, min-child-weight 6, subsample 0.85, column sample 0.8, $\ell_1 = 0.1$, $\ell_2 = 2.0$, seed 42, native NaN handling, and no early stopping. The floor classifier is a shallower

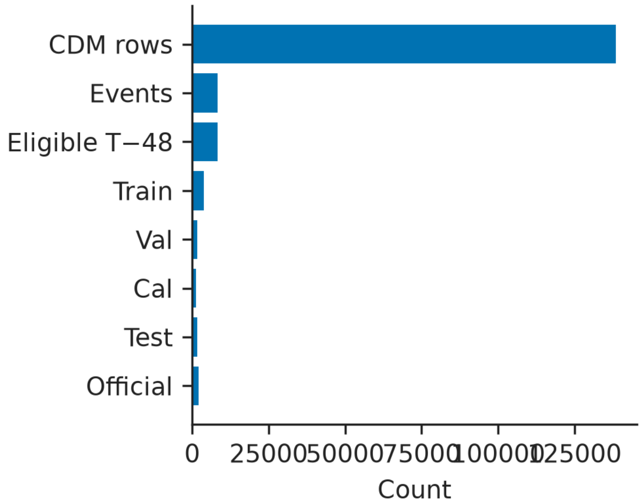

Fig. 1. Archive construction. 162,634 CDM rows yield 8,293 cutoff-safe events at T−48, then an event-disjoint split and a 2167-event official test.

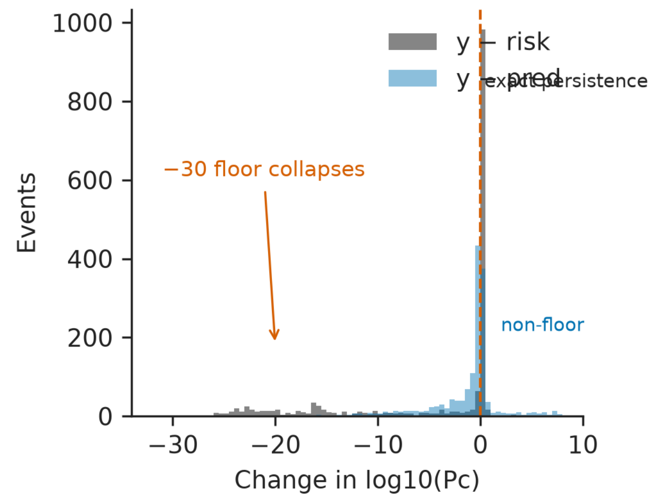

Fig. 2. Later movement $y-$ risk versus unguarded XGBoost residual. The spike is exact persistence; the left tail is collapse to -30.

booster (180 trees, depth 3) with train-set class weighting. Inference is CPU-only.

The live policy is a two-part hurdle: $P(y=-30)$, a residual booster fit on non-floor training events, threshold 0.15 chosen on validation, and no -6 persistence guard. Live intervals are split conformal around that point [10]. SHAP explains the residual regressor. Code: https://github.com/ArjunCodess/PRISM.

III. Structure of the Target

Of 1659 local-test events, 1352 later reports sit at -30 (floor rate 0.815). Persistence already has median absolute error 0: 55.6% of reports do not move after 48 hours. Fig. 2 shows the spike at $y-\text{risk}=0$ and a left tail of floor collapses. Treating the floor as ≤ -20 or ≤ -25 instead of -30 raises the floor rate to 0.845 and 0.828 and still leaves floor-excluded XGBoost MAE worse than persistence. A Tobit-style censored regression is a natural alternative formulation; this study uses a two-part hurdle instead of fitting a Tobit.

Unguarded XGBoost has 48.8% of test events within 0.5 log units and p_{90} absolute error 9.51. Mean error is the wrong headline for this distribution.

IV. Forecasting Experiments

TABLE I
Frozen local test at 48 h ($n=1659$; 1352 floor). MAE in $\log_{10} P_c$. Δ is MAE(persistence)−MAE(model) with a 95% bootstrap interval.

System	MAE	Med.	Non-fl.	Floor	Δ [95%]	p
Persistence	5.080	0.000	4.073	5.308	—	—
Trend	5.048	—	4.331	5.210	—	—
Unguarded XGB	2.809	0.554	7.562	1.730	2.270 [1.930, 2.638]	0.91
Residual XGB	2.760	0.453	7.375	1.712	2.319 [1.977, 2.685]	0.92
Floor hurdle	2.109	0.000	9.311	0.474	2.971 [2.524, 3.415]	$< 10^{-33}$
Guarded ens.	3.059	0.473	7.332	2.088	2.021 [1.716, 2.380]	0.31

Table I reports the frozen local test. Unguarded XGBoost lowers MAE from 5.080 to 2.809 ($\Delta=2.270$ $[1.930, 2.638]$) while floor-excluded MAE rises from 4.073 to 7.562. Wilcoxon on paired absolute error does not reject equality ($p=0.91$). Residual reconstruction is almost the same (MAE 2.760). The floor hurdle reaches MAE 2.109 and median AE 0 because it calls the floor, including on 174 non-floor events; floor-excluded MAE is 9.311. $F_2=0$ for the hurdle, residual model, and unguarded XGBoost. ESA-style loss still ties at 0.167 for persistence and an 18 August ensemble snapshot that copies today's report once that report is already ≥ -6.

A clipped one-step trend from the last two cutoff-safe reports barely moves MAE (5.048). Snapshot-only XGBoost is 2.911; history-only 2.977; covariance-only 3.148. Adding history summaries to the snapshot lowers MAE from 2.904 to 2.851; covariance trends reach 2.808. Across five grouped redraws, unguarded XGBoost MAE advantage is 2.34 ± 0.04 and floor-excluded MAE is 7.78 ± 0.21. Table II states the pattern without a leaderboard.

The floor classifier at the validation threshold 0.15 has recall 0.976, precision 0.883, AUROC 0.920, AUPRC 0.981, and Brier 0.114 (Fig. 3). The 174 false positives have median later $y=-14.3$; only 25% sit below -20. They are near-floor reports, not high-risk misses. A logistic model on risk alone already reaches AUROC 0.799; adding `max_risk_estimate` or using the gap alone stays near 0.80. XGBoost is better, but the floor phenomenon is visible to a two-feature logistic. Fig. 4 shows why: many later reports sit on the -30 line.

V. Forecast Horizon

Table III uses the *same* 1459 test events that have valid observations at 72, 48, 24, and 12 hours (7340 events in the

TABLE II
WHAT DOES NOT WORK, RELATIVE TO PERSISTENCE. OFFICIAL L IS URIOT ESA-STYLE LOSS ON THE FROZEN OFFICIAL TEST.

Approach	Mean MAE	Non-floor MAE	Official L
Persistence	baseline	baseline	strong
Trend	no	no	—
Unguarded XGB	yes	no	no
Residual XGB	yes	no	no
Floor hurdle	yes	no	no
Guarded ens.	some	no	ties

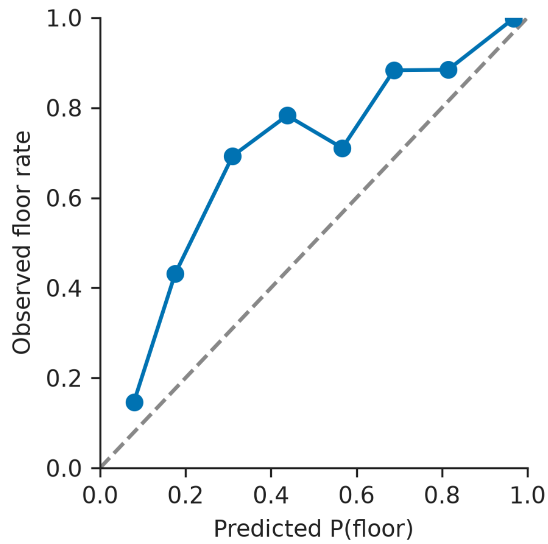

Fig. 3. Reliability of $P(\text{floor})$ for the hurdle classifier.

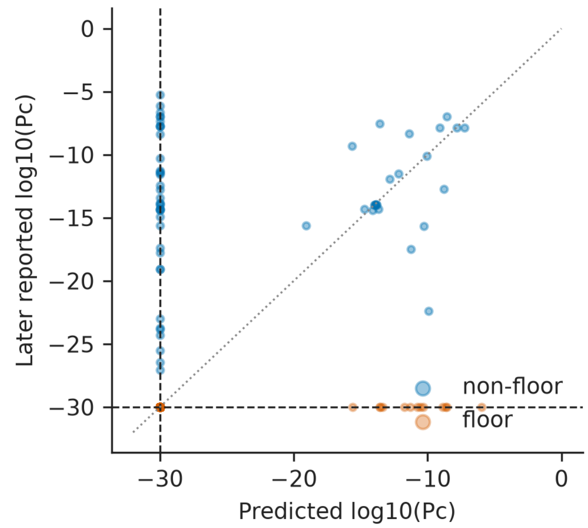

Fig. 4. Floor-hurdle forecast versus later reported $\log_{10} P_c$. The dashed line is the archive floor.

TABLE III
MATCHED-COHORT XGBOOST VERSUS PERSISTENCE. SAME EVENTS AT EVERY HORIZON ($n_{\text{test}} = 1459$).

h	Floor	Overall Δ	[95%]	Non-fl. Δ
72	0.824	4.487	$[3.96, 5.01]$	-3.86
48	0.824	2.198	$[1.76, 2.59]$	-3.95
24	0.824	0.428	$[0.18, 0.65]$	-2.67
12	0.824	0.029	$[-0.15, 0.20]$	-1.70

archive intersection). Target floor rate is therefore constant (0.824). Extra MAE beyond persistence still falls from 4.49 log units at 72 h to 0.03 at 12 h, where the bootstrap interval covers zero. Non-floor ΔMAE is negative at every horizon: the learned model is worse than persistence off the floor. Fig. 5 shows MAE with those intervals. The older overlapping-set table (advantage 4.53, 2.27, 0.52, 0.06) mixed a changing eligible population with time-to-approach; the matched cohort isolates the horizon effect.

VI. HIGH-RISK EVALUATION

The ESA class is $\log_{10} P_c \geq -6$ [3]. The local test has nine such events. Leave-one-out residual XGBoost on the 66 eligible high-risk events loses to persistence $66/66$ (MAE 12.59 versus 1.21). That is clean and statistically fragile. Under this dataset and protocol, these models did not demonstrate improved tail performance. They do not license the broader claim that machine learning cannot forecast high-risk conjunctions.

A mission-grouped 80/20 split (15 vs 4 missions) has 37 held-out high-risk events: overall XGBoost MAE 3.07 versus persistence 4.74, but high-risk MAE 11.9 versus 1.50. An earlier four-mission draw contained a single high-risk event (MAE 19.2) and is not interpretable for the tail. The public archive cannot test object-pair leakage.

Table IV is confirmation, not discovery. Persistence $L = 0.694$ matches Uriot last-risk-prediction. The published challenge entry *sesc* reached $L = 0.556$ [3]. Unguarded XGBoost has $F_2 = 0$ and $L = 1.75 \times 10^6$: one wildly overconfident miss on a real high-risk event blows up the numerator while the denominator is zero. The floor hurdle lowers MAE and raises L to 70.5. Fig. 6 annotates that bar. A class-cut sweep on frozen local predictions shows that this study cannot currently say whether the model helps at the tighter operational cuts near -4 to -5; those cells have $n+ \leq 3$ and $F_2 = 0$ for the learned systems.

VII. COVARIANCE AND THE MAX-RISK GAP

Combined snapshot $\log\det$ correlates with later $|y - \text{risk}|$ ($\rho = 0.399$ $[0.36, 0.44]$). The max-risk gap correlates with ending at the floor (logistic test AUC 0.819) and with *less* later movement ($\rho = -0.768$ $[-0.79, -0.75]$). After standardizing and partialling risk, n_{messages}, and miss distance, the $\log\det$ coefficient on $|y - \text{risk}|$ is -0.60: the univariate covariance association is not an independent partial effect once today's report is in the model. Association, not cause. Fig. 7 is the scatter; quartile tables remain in the supplement. Residual SHAP is dominated by risk-trajectory fields (`max_risk_scaling`, risk), not covariance volume.

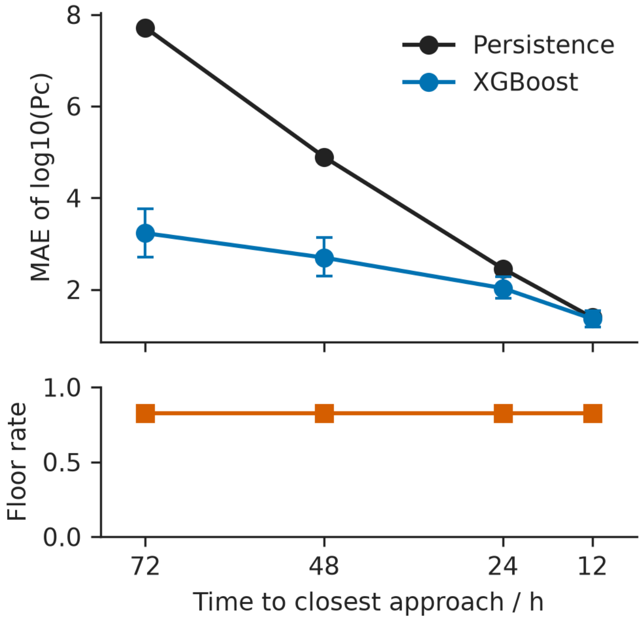

Fig. 5. Matched-cohort MAE versus time to closest approach, with floor rate on the same events.

TABLE IV
OFFICIAL TEST ($n = 2167$; 150 HIGH-RISK; 1673 FLOOR). FROZEN BEFORE LOOK. $L = \mathrm{MSE}_{HR}/F_2$.

System	MAE	Med.	Non-fl.	L	F_2
Persistence (LRP)	5.209	0.000	4.287	0.694	0.739
Unguarded XGB	3.504	0.690	9.333	1.75×10^6	0.000
Residual XGB	3.476	0.594	9.289	104	0.017
Floor hurdle	3.177	0.000	11.832	70.5	0.025
Guarded ens.	3.107	0.439	6.750	0.694	0.739

VIII. UNCERTAINTY AND ABSTENTION

Bootstrap 90% spread covers 47.7% of the local test. Split conformal around the floor-hurdle point covers 90.1% $[0.887, 0.914]$ with constant width 21.03 log units. Floor events are well covered (97.6%); non-floor coverage is 57.0%. Calibrated uncertainty is statistically honest and operationally almost useless at this heterogeneity. Abstaining when the 90% band crosses -6 accepts 90.4% of events (159 to review), accepted MAE 1.706, and false reassurance 2 of 9. Tighter conformal radii do not abstain until the 90% quantile, because most calibration residuals are exact floor hits (Fig. 8).

IX. DISCUSSION

Use this model, if at all, to triage reports that are likely to collapse to the archive floor. Do not use it to assess events that are already flagged high-risk. Selecting on MAE ships a floor-calling policy. Selecting on ESA-style loss would have kept persistence or the guard. Persistence becomes strong near closest approach because later reports have already absorbed later tracking; the matched-cohort decay is the statistical counterpart of that absorption.

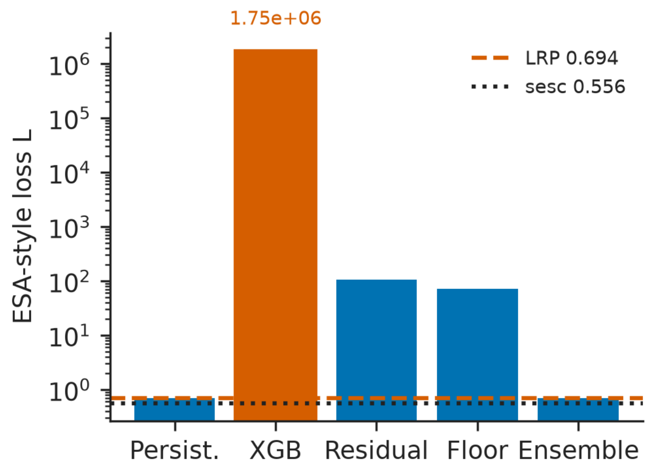

Fig. 6. Official-test ESA-style loss. Persistence matches Uriot LRP 0.694; sesc is 0.556. The unguarded bar is annotated because it is not a typo.

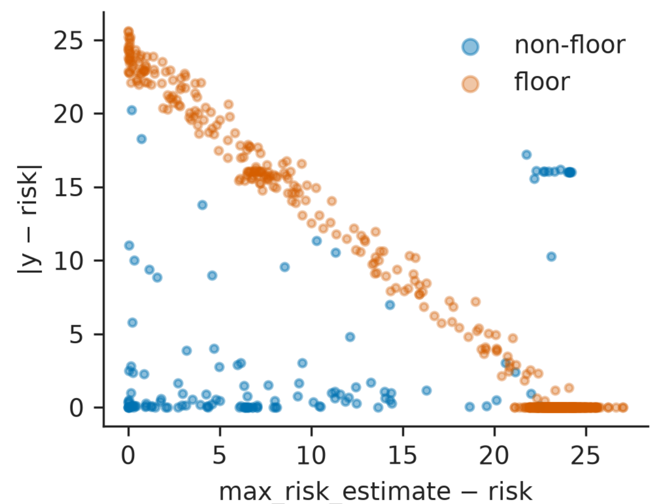

Fig. 7. Max-risk gap versus later $|y - \mathrm{risk}|$. Floor events cluster at large gap and small movement.

The 2015–2019 archive predates today's denser catalogues [5], [6]. Floor-collapse rates could change with tracking cadence and mega-constellation geometry. Manoeuvres are out of scope. The target is later reported $\log_{10} P_c$, not physical collision probability. H1–H4 were pre-specified; sweeps are exploratory. Code, frozen split IDs, hyperparameters, and figure scripts live at https://github.com/ArjunCodess/PRISM; data remain ESA/Zenodo 4463683. This is a research prototype, not a warning system.

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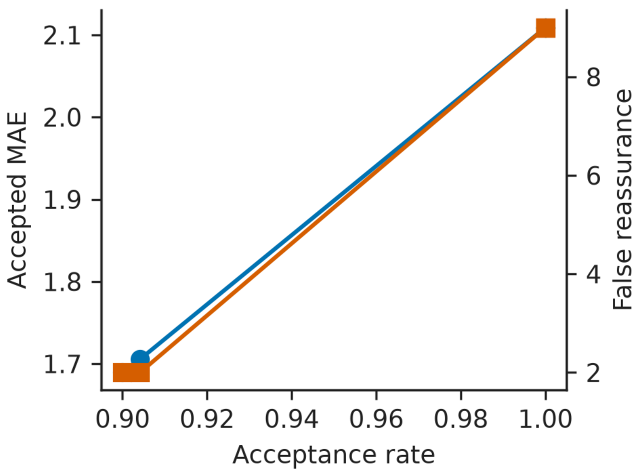

Fig. 8. Selective prediction under the live conformal rule as nominal coverage changes.

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